

Extrusion Cheat Sheet

Theoretical

Based on a modified Ender syringe extruder. Designed for a 2 mm nozzle but easily adapted. A practical starting point for calibrating and printing pastes, gels, inks, clays and similar materials.

0 Before you're able to calibrate the extrusion while printing, you have to **calibrate the E-steps**. These dictate how much the extrusion motor needs to turn to extrude 1 mm of filament.

Start by inputting the theoretical value in the Gcode. Then fine-tune this value by commanding 50 mm extrusion and then measure the actual length.

$$E_{\text{steps}} = \frac{3.0625 * S_M * M * R}{D^2 * p}$$

S_M : motor steps per rev. (typical 200)

M : microsteps (typical 16)

R : gear reduction ratio

D : syringe inner diameter (mm)

p : lead screw pitch (mm/rev.)

Fine-tune

$$E_{\text{new}} = E_{\text{old}} * \frac{\text{commanded } E}{\text{actual } E}$$

Ender 3-V2 printed

70

5 The **start-stop test** can reveal the printer's and material's ability to deal with priming and retraction.

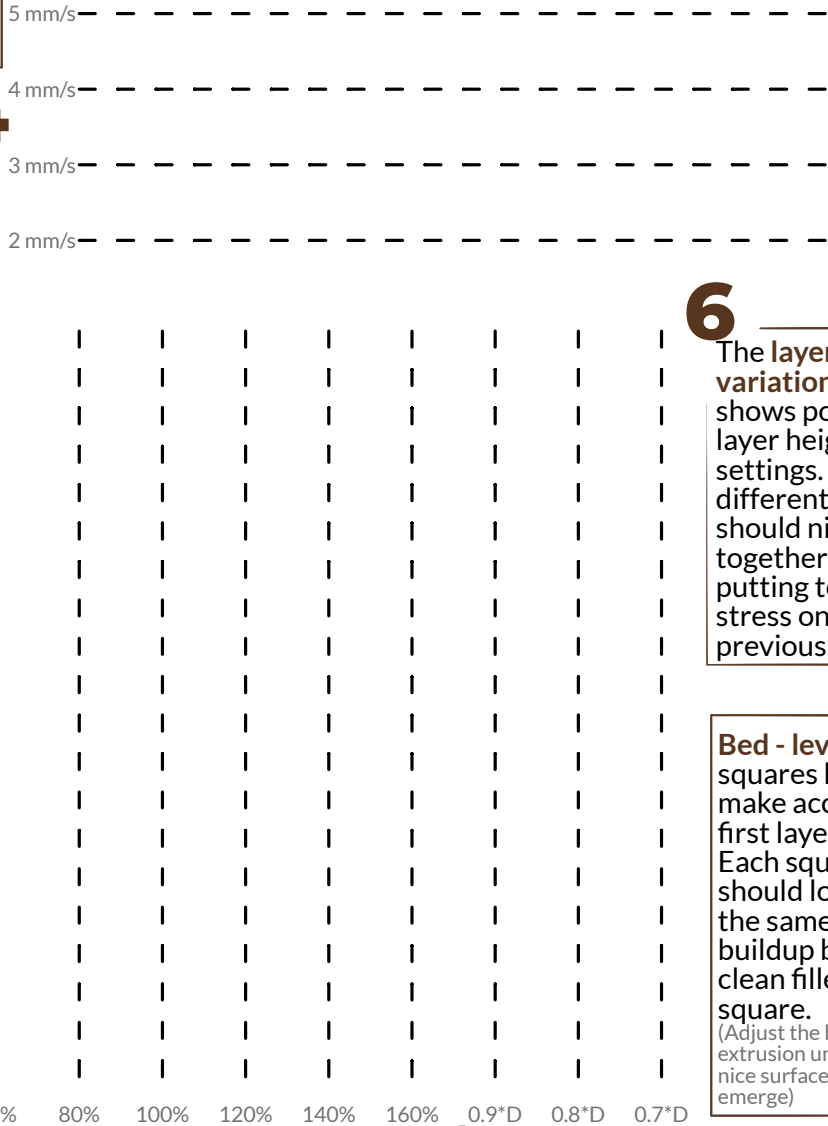
4 Use the best extrusion factor and examine the **printing speed** limits for consistent extrusion.

6 The **layer height variation** squares show possible layer height settings. The different layers should nicely fuse together without putting too much stress on the previous.

3 **Bed - leveling** squares help make accurate first layers. Each square should look the same, no buildup but a clean filled square. (Adjust the bed and extrusion until four nice surfaces emerge)

2 Use the **layer height** that gives the most consistent and closest to the nozzle diameter width.

1 Measure the width of the different **extrusion factors**. Use the most consistent and/or slightly smaller than the nozzle's diameter. (at 1 mm/s)



Layer Height
1.0 x dia Nozzle

Layer Height
0.6 x dia Nozzle

Layer Height
1.2 x dia Nozzle

Layer Height
0.8 x dia Nozzle